PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

ARUSHA REGION FORM IV PRE-MOCK EXAMINATION

032 – CHEMISTRY

MARKING SCHEME. F.4, MARCH. 2026

SECTION A

Multiple choices (10 marks)

1

i	ii	iii	iv	v	vi	vii	viii	ix	x
A	B	D	A	C	B	C	C	B	E

(1 mark @ = 10 marks)

Matching items (6 marks)

2

LIST A	i	ii	iii	iv	v	vi
LIST B	H	F	E	B	G	C

(1 mark @ = 06 marks)

3. (a) (i) A base is the substance that neutralizes an acid by reacting with hydrogen ions while Alkali are bases which are soluble in water (1 mark)

(ii) An atom is the smallest particle of an element which take part in chemical reaction while Isotopes are atoms of the same element that contain the same number of protons but different number of neutrons in the nucleus. (1 mark).

3 (b). Solution

Step 1.

Element/symbol	C	H	O
Percentage composition %	52.2	13	34.18
% composition ÷ R.A.M	$\frac{52.2}{12} = 4.35$	$\frac{13}{1} = 13$	$\frac{34.18}{16} = 2.136$
Divide by the lowest value both sides	$\frac{4.35}{2.136}$	$\frac{13}{2.136}$	$\frac{2.136}{2.136}$
simple ratio	2	6	1

The tempirical formula is C_2H_6O (1mark)

Step 2: Determine the molecular formula.

$Mr = 2 \times VD = 2 \times 23 = 46 g/mol$ (1mark)

Molecular Formula = n (sum of relative atomic masses in C_2H_6O)

But RMM = n (tempirical formula)

$46 = n((2\times12) + (6\times1) + (1\times16))$ (1mark)

$\frac{46}{46} = \frac{46n}{46}$

$n = 1$.

The molecular formula is $n \times (C_2H_6O)$, but $n = 1$.

$1 \times (C_2H_6O)$

∴ The molecular formula is C_2H_6O (2 marks)

(c) The oxidation state of sodium in sodium sulphate.

3 (c) Soln

The oxidation state of Na in Na_2SO_4.

$2Na + -2 = 0$

$\frac{2Na}{2} = \frac{+2}{2}$

$Na = +1$ (2 marks)

4 (a) (i) Electrolyte is copper (II) sulphate solution ($CuSO_4$). (01 mark)

Anode – Blister copper (00½)

Cathode – pure copper (00½)

(ii) Observation: Copper is deposited on the cathode sheet. As the electrolysis continues, the whole blister copper dissolves which eventually increases the size (mass) of the pure copper plate (02 marks)

4 (b) (i) Element A is sulphur (01 mark)

catalyst B is vanadium pentaoxide (V_2O_5) (01 mark)

An acid C is sulphuric acid (H_2SO_4) (01 mark)

(ii) Equation for the formation of sulphur dioxide in stage 2 is

$$2SO_{2(g)} + O_{2(g)} \xrightarrow[450^oC\ 2\text{-}3atm]{V_2O_5} 2SO_{3(g)}$$

(02 marks)

5 (a) Temporary hardness of water is the water contains dissolved salts of calcium hydrogen carbonate or magnesium hydrogen carbonate WHILE Permanent hardness of water contains dissolved salts of calcium sulphate or magnesium sulphate.

(03 marks)

(b) Temporary hardness of water can be soften by boiling

Equation:

$$Ca(HCO_3)_2 \xrightarrow{\Delta} CO_{2(g)} + H_2O_{(l)} + CaCO_{3(s)}$$

(02 marks)

- Permanent hardness of water can be soften by adding some chemicals ie Na_2CO_3.

$$Na_2CO_{3(aq)} + CaSO_{4(aq)} \longrightarrow CaCO_{3(s)} + Na_2SO_{4(aq)}$$

$$Na_2CO_3 + Ca(HCO_3)_{2(aq)} \longrightarrow CaCO_{3(s)} + NaHCO_{3(aq)}$$

Temporary hardness of water (02 marks)

(c) (i) Sample D and A (01 mark)

(ii) sample A - forms lather after boiling
- Its hardness was removed after boiling
(01 mark)

6 (a) (i) standard solution is a solution with a precisely known concentration of a substance

(ii) Equivalent point of titration is a point whereby moles of titrant equals to the moles of analytes in the solution

- The point when moles of added titrant are stoichiometrically equal the moles or analytes in the solution which resulting in complete reaction.

(iii) Basicity of an acid – is the number of ionisable hydrogen atoms per molecule of an acid that can be displaced by a metal in solution.

(01 @ = 03marks)

(b) (i) Balanced chemical equation

$$XOH_{(aq)} + HCl_{(aq)} \longrightarrow XCl_{(aq)} + H_2O_{(l)}$$ (01 mark)

(ii) The molarity of XOH solution

From

$$\frac{M_a V_a}{M_b V_b} = \frac{n_a}{n_b}$$

volume of acid, $V_a = 35\ cm^3$
volume of base, $V_b = 25\ cm^3$
Molarity of acid, $M_a = 0.1M$
Molarity of base, $M_b = ?$

6 (b) (ii)

$$M_b = \frac{M_a V_a n_b}{V_a n_a} \quad \text{— (01 mark)}$$

$$= \frac{0.1M \times 35\,cm^3 \times 1}{25\,cm^3 \times 1}$$

$$M_b = 0.14M$$

∴ Molarity of the metal hydroxide is 0.14 M (01 mark)

(iii) The relative molecular mass of XOH ?

Given $25\,cm^3$ of XOH = 0.196 g

$1000\,cm^3$ of XOH = ?

$$= \frac{1000 \times 0.196}{25} = 7.84\,g/dm^3 \quad \text{(01 mark)}$$

$$\therefore \text{Molar mass} = \frac{\text{Conc}\,(g/dm^3)}{\text{Molarity}} = \frac{7.84\,g/dm^3}{0.14\,mol/dm^3} = 56\,g/mol$$

(01 mark)

∴ Relative molecular mass of XOH is 56

(01 mark)

XOH = 56

X + 16 + 1 = 56

X + 17 = 56 − 17

X = 39

∴ The relative molecular mass of X is 39.

(01 mark)

07 (a) Differences between homogenous mixture and heterogenous mixtures

Homogenous mixtures	Heterogenous mixtures
- Have a uniform composition, appearance and properties	- Have different composition appearance and properties at various points in the mixture.
- After stirring, every section of the solution is identical in composition, appearance and physical properties	- Contains substances that are not evenly distributed for example Sand and water
- An example is small amount of common salt dissolved in a glass of water	- An example when sulphur when sulphur powder and iron filling are mixed forms heterogenous mixture
Uniformly distributed	- Not uniformly mixed and particles settle on standing at the bottom.

(01 @ = 04 marks)

(b) Common salt is a compound because

(i) Sodium and chlorine are chemically combined in fixed ratio

(ii) The combination of sodium and chlorine can be separated by physical means

(iii) When salt is formed there is a change in energy or mass.

(iv) The properties of salt is not common to its constituent elements 7.

7 (b) (iv) OR.

The properties of salt is different from its constituents

(v) There is a change in composition of constituents when the common salt is formed from its elements.

(01 @ = 05 marks)

8 (a) (i) Molecular formula is a chemical formula that shows the total number of atoms of each element in a molecule.

(ii) Cracking is the breaking of larger hydrocarbon molecules into smaller hydrocarbon molecules.

(iii) Homologous series is a series of organic compounds with the same functional group and similar chemical properties, in which the members differ from each other by $-CH_2$ group.

(iv) Isomers are compounds with the same molecular formula but different structural formulae.

(01 mark @ = 04 marks)

8 (b)

C_7H_{12} — alkyne

C_4H_{10} — alkane

C_5H_{10} — Alkene

C_3H_4 — Alkyne

C_6H_{12} — Alkene

SECTION C (30 marks)

Two questions only

9 Introduction — (01½), conclution (01½)

Causes of water pollution

- Using pesticides near water sources
- Using fertilizers near water sources
- oil spills on the surface of water
- mining - produce waste
- sediments
- Industrial proces
- sewages
- poisonous gases and metals eg SO_2, N_2 NH_3 and mercury

(04 marks)

Effects

- infectious diseases
- Excessive growth of water weeds
- chemical spills and leak into water bodies kill acquatic animals
- Rise in water temperature
- Cause lung, nasal and sinus cancers eg chromium (VI)
- Effect on human health especially in young children

(04 marks)

9 Remedial measures
- Reduce the use of fertilizers and pesticides
- treating sewage and industrial wastes
- Stopping deforestation
- Controlling coastal development
- Reducing pollution from oil spills
- Reducing the use of mercury
- Cleaning our existing and abandoned mines
- Cleaning up chemical pollutants
- Banning the use of lead pipes

10 Introduction (01½)
main body (02@ - 12 marks)
conclusion (01½)

(i) In production of gases
eg Cl_2, H_2 and oxygen
(ii) In purification of copper
(iii) Electroplating of metallic materials
(iv) Extraction of metals
(v) Production of chemicals eg sodium hydroxide from brine
(vi) Medical electrolysis eg Permanent removal of hair follicles

10 (VII) Production of heavy water eg manufacture of denterium oxide (D_2O)
(VIII) Surface cleaning and finishing
(IX) Production of hydrogen fuels

11 Introduction (01½)
main body (02 @ = 12 marks)
conclusion (01½)

(I) Prevention of soil erosion
(II) Crop rotation
(III) Good harvesting practices
(IV) Controlling grazing
(V) Mixed cropping (inter cropping)
(VI) Use of manures
(VII) Agroforestry
(VIII) Minimum tillages
(IX) Mulching